

AI Audit Report (Mandatory Appendix)

Declaration: *I use AI tools for the following tasks.*

- **AI tool:** Claude (Opus 4.8) via Claude Code CLI.
- **Session date:** 2026-08-30.
- **Role of AI:** designed and generated the JMeter test plans, the CSV data, the analyzer and runner scripts, the CPT proposal, and the agent skill; drove test execution; analysed the raw `.jtl` logs. **All output was reviewed and corrected by the student** (see Main Report §3).

Human-only work (NOT AI-generated, per anti-cheat §11): the demo video + narration, the resource-monitor screenshots, the `dxdiag` hardware report, and the GitHub issue filings.

Interaction log

[1] 2026-08-30 — Master plan + endpoint mapping

- **Prompt:** "read the HW05 requirements ... the previous features I chose in hw2/test-cases. Create a master plan step by step and guide me to install things."
- **AI output (summary):** Read the requirements + HW02 test cases; mapped FR20/FR05-15/FR07-09 to auth/read/transactional; produced `MASTER-PLAN.md`; guided JMeter install and SUT startup.
- **Human review:** Accepted mapping; confirmed tool = JMeter, SUT already local.

[2] 2026-08-30 — Setup (install + run)

- **Prompt:** "installing and running the backend for me."
- **AI output:** Installed backend deps, seeded DB, started `node server.js` (port 3000), downloaded + unzipped JMeter 5.6.3, verified login + products endpoints, registered a 50-user pool via `POST /api/register`.
- **Human review:** Verified server reachable; confirmed seed accounts.

[3] 2026-08-30 — Generate + validate test plans

- **Prompt:** "continue full workflow, make sure commit step by step."
- **AI output:** Wrote `generate_plans.js`; produced Load/Stress/Spike `.jmx`; smoke-tested the workflow (all 6 samplers HTTP 200); parameterized threads via `${__P(...)}`; committed each plan separately.
- **AI mistakes caught (Main Report §3):** wrong coupon path; missing `user_id` extraction; single user vs pool; missing auth header; averages-only. All corrected before final plans.

[4] 2026-08-30 — Execute + generate reports

- **AI output:** `run_all.sh` ran Load/Stress/Spike/Soak headless, produced raw `.jtl` + HTML dashboards under `results/`.

[5] 2026-08-30 — Analyse logs (Task 2)

- **AI output:** `analyze_jtl.js` computed p50/p95/p99/throughput/error% from the raw logs; AI drew conclusions; student verified each number against the raw `.jtl` (see AI-Analysis-Critique.md).

[6] 2026-08-30 — CPT proposal + agent skill

- **AI output:** `CPT-Proposal.md` (flow chart + trade-offs); `performance-testing` skill + refs.

Verbatim prompts are the four user messages above; full assistant outputs are the committed artifacts in this repository (see `git_commit_log.txt` for the step-by-step trail).